

SHANMUGARAJ T

Data Analyst | SQL · Python · Power BI

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PROFESSIONAL SUMMARY

Data Analyst with 2.5+ years of experience in large-scale data operations, ETL, KPI reporting, and Business Intelligence dashboard development across SQL, Python, Excel, and Power BI. Proven track record building end-to-end data analytics pipelines, from database extraction and cleaning to categorization, dashboard development, and campaign reporting on lender and borrower profiles. Actively building toward fraud and risk analytics through hands-on fintech data segmentation work and self-directed projects in entity resolution and financial dashboarding.

WORK EXPERIENCE

Operations Executive — Data Operations & Reporting

June 2025 - Present

Armsotech Pvt Ltd

- Managed and processed large-scale election datasets across 100+ constituencies for 3 major clients, performing data validation and integrity checks to ensure accurate, timely delivery.
- Led a 4-member data operations team, coordinating task allocation, quality review, and delivery timelines across concurrent client projects.
- Partnered directly with client stakeholders across 3 concurrent projects to gather business requirements, define reporting scope, and align data delivery timelines (stakeholder management).
- Performed data extraction, transformation, and structuring (ETL) using SQL to enable efficient analysis, reporting, and downstream risk/quality checks.
- Designed structured relational data models linking voter, street-master, and lender/borrower datasets (data modeling) to support consistent, scalable reporting across projects.
- Conducted root cause analysis on data discrepancies identified during validation cycles, tracing errors back to source data and resolving them prior to client delivery — strengthening overall data quality and data integrity.
- Designed structured Excel workflows for data cleaning, validation, and team distribution, reducing errors and improving turnaround.
- Automated recurring reports using Excel VBA, significantly reducing manual effort and processing time.
- Developed Power BI dashboards and KPI reports to track weekly and monthly survey trends, supporting Business Intelligence needs across client teams.
- Built Python-based web scrapers with AI-assisted extraction logic and Selenium automation to collect and structure large volumes of data, cutting manual collection time significantly.
- Developed an OCR pipeline using Python and Tesseract to extract and validate structured ID data from scanned documents, with OpenCV-based image preprocessing for accuracy.
- Used openpyxl to automate multi-sheet Excel report generation and formatting across recurring data pipelines.
- Led migration of manual survey data collection to a mobile-based system (data migration), improving data capture efficiency and reducing processing time by 55%.
- Project — SmartPaiseCare (Fintech): Built a data pipeline for a loan-servicing campaign system, extracting borrower and lender data from MySQL and categorizing accounts by lender profile and repayment status (due, overdue, alert-tier).
- Performed exploratory data analysis (EDA) on lender and borrower datasets to identify usage and repayment trends, informing risk-tier segmentation logic.
- Automated segmentation logic in Python to classify loan accounts by risk tier and due status, generating structured reports that fed SMS campaign triggers (payment due alerts, overdue reminders, loan offer messaging).
- Used Excel and MySQL to validate and structure lender/borrower datasets prior to campaign dispatch, improving targeting accuracy and reducing message errors.

- Owned the end-to-end flow from database extraction through segmentation, reporting, and campaign trigger — direct exposure to loan lifecycle and borrower risk-tier data.

Web Analyst

Sep 2023 - June 2025

Ninestars Information Technologies Pvt Ltd

- Optimized websites for 250+ clients, improving user engagement, load performance, and overall site efficiency.
- Extracted and processed data from 100,000+ articles using Python, pandas, and XPath, increasing data retrieval speed by 30%.
- Built Selenium-based automation scripts to handle dynamic, JavaScript-rendered pages during large-scale data extraction.
- Utilized SQL for efficient data extraction and query optimization to support accurate analysis and reporting.
- Used pandas for data cleaning, transformation, and structuring of extracted datasets prior to reporting.
- Built Excel dashboards with VBA automation, reducing reporting time by 25% and improving KPI visibility.
- Applied data storytelling techniques to present insights clearly and support data-driven business decisions.

US IT Recruiter

Feb 2022 - Oct 2022

KRG Technologies

- Experienced in US IT recruitment with strong knowledge of visas, work laws, and employee benefits.
- Skilled in end-to-end hiring, including sourcing, screening, interviewing, and documentation.

PROJECTS

Finance Dashboard — Power BI

- Built an interactive Power BI dashboard for financial reporting, using DAX measures and Power Query to consolidate and visualize income, expense, and margin trends for data-driven review.

Electoral Data Matching Engine — Python (KYC-style Entity Resolution)

- Designed a Python-based record-matching pipeline to reconcile voter records against a reference street/address master using fuzzy and rule-based matching — structurally similar to KYC-style entity resolution (name/ID/address matching across data sources).
- Built automated OCR extraction (Tesseract, OpenCV, pdf2image) with regex-based validation to digitize and verify structured ID data at scale.

Voter Data Automation Suite — Python, Selenium, Flask

- Built a Flask-based web interface and Selenium automation tools for data upload, validation, and portal submission workflows, replacing manual, error-prone processes.

EDUCATION

Bachelor of Engineering (B.E.) — Mechanical Engineering

New Prince Shri Bhavani College of Engineering and Technology

CERTIFICATION

Data Analytics & Business Intelligence — Besant Technologies, April 2025

SKILLS

- Languages & Tools: Python, SQL (MySQL), Power BI (DAX, Power Query), Excel (VBA, Macros, VLOOKUP), openpyxl
- Python Libraries: pandas, Selenium, OpenCV, Tesseract (OCR), pdf2image, XPath
- Data & Analytics: Data Cleaning, Data Validation, Data Segmentation, Entity/Record Matching, Data Visualization, Dashboarding, Reporting, Process Automation
- Domain Exposure: Loan/lender data categorization, borrower risk-tier segmentation, campaign reporting (fintech)
- Core: Analytical Thinking, Team Leadership, Communication, Problem-Solving